

§3.1

Numbers in Other Bases

One place value system for every base, and two ways to move between them

Today's Goal

After completing this section, you should be able to:

- Express and read a number written in a base other than ten using the place value system for that base
- Convert a number from base ten to a given base b , and from base b back to base ten
- Use the Exploding Dots model to represent numbers and to convert between base ten and other bases

Learning Target 9

LT.9 Different Number Bases

I can convert numbers from base ten to base n and vice versa for any whole number $n > 1$. I can add, subtract, multiply, and divide in different number bases.

Why Any Base But Ten

Who needs a base other than ten?

Link shorteners do. That short code is a number written in base 36.

Base 36 uses 0 to 9, then a to z. So a is 10 and z is 35.

Link number 12,345 becomes the code **9ix**. Five digits, then three.

One Place Value System per Base

Definition 3.1.1

Let b be a whole number greater than 1, called the *base*. In the *base- b place value system*, the positions of a numeral (read from right to left) have place values

$$1, \quad b, \quad b^2, \quad b^3, \quad \dots,$$

so the position n places from the right has place value b^{n-1} . A numeral written in base b , denoted N_b (read “ N in base b ”), has value equal to the sum of each digit multiplied by the place value of its position.

Four Ways to Write It Down

When the base is not ten we always say so. For base 5, any of these:

- the base 5 number 265
- 265_{five}
- $(265)_5$
- $(2, 6, 5)_5$

How Not to Say It

Watch out

We read 265_{five} as “two-six-five in base 5”, never as “two hundred sixty-five in base 5”. Base 5 has no hundreds place and no tens place.

What a Numeral Is Worth

Determine the value of 3042_{five} .

Position	Place Value	Digit	Value of Digit
1st	1	2	$2 \cdot 1 = 2$
2nd	5	4	$4 \cdot 5 = 20$
3rd	$5^2 = 25$	0	$0 \cdot 25 = 0$
4th	$5^3 = 125$	3	$3 \cdot 125 = 375$

Add the last column: $375 + 0 + 20 + 2 = 397$.

Which One Is Bigger?

1000_{two} **or** 100_{three} ?

$$1000_{\text{two}} \text{ is } 1 \cdot 2^3 = 8.$$

$$100_{\text{three}} \text{ is } 1 \cdot 3^2 = 9.$$

So 100_{three} is bigger, even with fewer digits.

Going the Other Direction

Write the base ten number 500 in base 6.

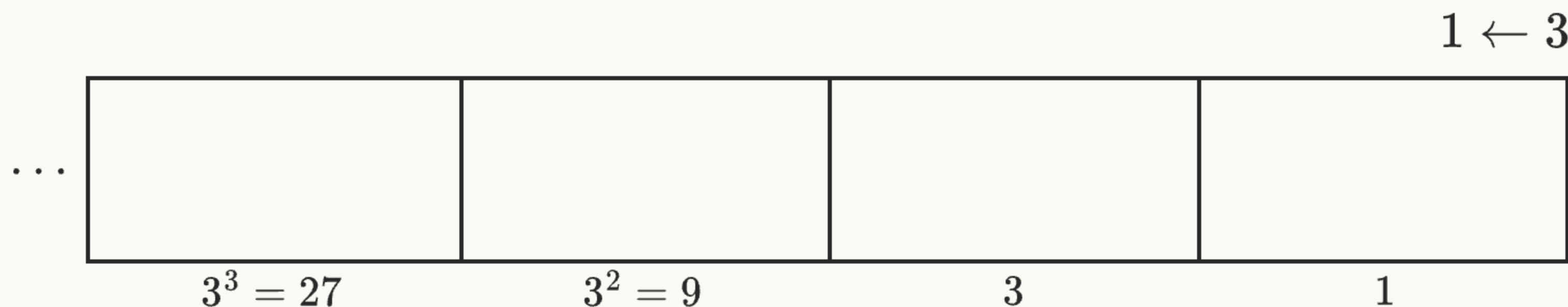
Base 6 places: 1, 6, 36, 216, 1,296. Since $216 < 500 < 1,296$, the 216 place is the highest that fits, so four digits.

Step	Divide	Digit	What is left
1	$500 \div 216$	2	68
2	$68 \div 36$	1	32
3	$32 \div 6$	5	2
4	$2 \div 1$	2	0

Read the Digit column down: 2152_{six} .

A Machine for Regrouping

Same dots and boxes as §1.1, with one change: the base need not be ten.



Every 3 dots in a box regroup into 1 dot in the box to the left.

Counting by Explosions

Use the Exploding Dots model to write 11 in base 3.

$$1 \leftarrow 3$$

Step 1



11 dots

Step 2



3 groups explode

Step 3



1 group explodes

No groups of 3 remain, so read the boxes: 102_{three} .

Find the Base

**In base b , the numeral 24
is worth 14. What is b ?**

The 4 sits in the ones place. The 2 sits in the b place.

So $2b + 4 = 14$, giving $b = 5$. Check: $2 \cdot 5 + 4 = 14$.

One Question to Leave With

The bigger the base, the shorter the numeral. Why?

Each place is worth b times the one to its right. Raise b and every place carries more, so fewer places are needed.